CHRISTIAN LIAN PAULO PEREZ RIOFLORIDO

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SUMMARY

Extensive experience developing quantum artificial intelligence and deep learning architectures for advanced spatiotemporal forecasting and autonomous robotics. Proven track record of building end-to-end forecasting pipelines on enterprise NVIDIA HPC clusters and cloud infrastructures, alongside deploying optimized computer vision and NLP models for real-world commercial applications. Led multidisciplinary research initiatives, secured industry-backed funding, published in top-tier academic journals, and mentored engineering teams across academic and corporate environments.

EDUCATION

Chung Yuan Christian University Taoyuan City, Taiwan
Ph.D. Electrical Engineering December 2026
Dissertation: Quantum Artificial Intelligence for Resilient Power Systems: Advanced Spatiotemporal Forecasting of Renewable Energy and Extreme Weather Events (Prof. Ying-Yi Hong)

Chung Yuan Christian University Taoyuan City, Taiwan
M.S. Electrical Engineering 2018
Thesis: A Hybrid Deep Learning-Based Network for 24-Hour Ahead Wind Power Forecasting (Prof. Ying-Yi Hong)

Adamson University Manila, Philippines
B.S. Electronics Engineering 2014
Thesis: Smoke Emission Meter Using Image Processing With Cloud Application(Engr. Mark Angelo Purio)

SKILLS

Programming & Tools: Python, C/C++, MATLAB, CUDA, Docker, Git, Linux shell, MySQL, CUDA-Q, PennyLane, Qiskit
Software: PyTorch, TensorFlow, Lightning, cuDNN, OpenCV, OpenVINO, ONVIF, Caffe, ONNX.
Languages: English, Filipino, Chinese

RESEARCH EXPERIENCE

Chung Yuan Christian University Taoyuan City, Taiwan
PhD Graduate Researcher September 2021 – Present

- NVIDIA Collaboration & Grant: Secured the NVIDIA Academic Grant for the project "Hybrid Deep Learning and Quantum Network for Typhoon Forecasting in a Wind Power Farm", published in Energy. Co-authored research with NVIDIA AI Technology Center on Hybrid Graph-Quantum Neural Networks, published in IEEE Transactions on Power Systems.
- Quantum & Spatiotemporal Forecasting: Designed quantum-inspired QNN/QLSTM and deep learning architectures (TFT, CNN-LSTM) for typhoon track/intensity and solar GHI forecasting.
- HPC & Acceleration: Built end-to-end forecasting pipelines using PyTorch/Lightning on NVIDIA A100/H100 clusters; accelerated energy workloads with cuQuantum and ran typhoon circuits exclusively on CUDA-Q.
- LLM Integration: Led experiments and ablations for LLM-based prediction workflows, ensuring reproducible evaluation standards.

MS Graduate Researcher 2016 – 2018

- Prototyped hybrid neural networks blending classical and deep models to leverage complementary biases for time series wind power forecasting, published in Applied Energy
- Performed extensive data analysis on wind turbine datasets to uncover temporal/seasonal drivers.

PROFESSIONAL EXPERIENCE

President Information Corp. Taipei City, Taiwan
Intern August 2023 – March 2024

- Inventory Optimization: Deployed a real-time object detection system using NVIDIA Metropolis, achieving a ~20% reduction in stockouts and boosting inventory accuracy by ~15%.
- Retail Analytics: Engineered automated CV models for shelf scanning and product recognition to drive dynamic price optimization and merchandising strategies.

- Customer Engagement: Implemented a facial recognition pipeline to enable personalized in-store services and targeted marketing campaigns.

New Era AI Robotics, Inc. (New Kinpo Group) New Taipei City, Taiwan
Deep Learning Software Engineer October 2018 – June 2021

- Computer Vision Deployment: Delivered production-grade models for object detection, tracking, and face recognition using PyTorch, TensorFlow, and Caffe.
- Multilingual NLP Development: Spearheaded the development of Speech-to-Text and QA models supporting Thai, Tagalog, Chinese, and English, integrating BERT-like LLMs in collaboration with NVIDIA.
- System Integration: Orchestrated communication between vision and speech nodes using ROS middleware to synchronize sensory inputs with robotic responses.
- Edge Integration: Integrated AI models with ONVIF intelligent cameras for security analytics and developed utilities for efficient data ingestion and MLOps workflows.

ROHM Electronics Philippines, Inc. Cavite, Philippines
Test Engineer May 2015 – August 2016

- Improved semiconductor production quality by investigating failures and refining test protocols/code to enhance signal processing and fault detection.

TEACHING EXPERIENCE

Chung Yuan Christian University Taoyuan City, Taiwan
Graduate Teaching Assistant Fall 2016, Fall 2021, Spring 2025–2026

- Courses: Artificial Intelligence; Microgrid; Machine Learning & Deep Learning; Big Data Analysis
- Delivered lectures, lab sessions, and project advising across multiple cohorts; taught deep learning for vision and forecasting to 25+ students per semester.

Adamson University Manila, Philippines
Invited Lecturer October 2026

- Course: Renewable Energy: Design and Control of Wind Energy Systems (Virtual / Online)
- Designed and delivered a multi-session instructional series focusing on the principles, design methodologies, and control strategies for wind energy systems.
- Conducted 3-hour intensive lecture blocks to structure the curriculum.

Technological Institute of the Philippines Quezon City, Philippines
Adjunct Professor/ Thesis Advisor August 2022 – May 2023

- Courses: Signal, Spectra, and Signal Processing; Prediction & Machine Learning
- Designed curriculum materials, homework assignments, projects, and student evaluation metrics.
- Supervised 5 undergraduate thesis teams, guiding problem formulation, experimental design, and defense. 3 projects led to peer-reviewed papers with students as first authors (IEEE TENCON 2023; IET-ICETA 2022 ×2).

LEADERSHIP AND SERVICE

National Taiwan University Quantum Club (NTUQC) Taipei City, Taiwan
Competitions Chair August 2026 – Present

- Direct and administer competitive programming and quantum problem-solving challenges for student members.
- Coordinate with academic and institutional partners to design competition tracks aligned with current quantum machine learning and quantum computing frameworks.

Association of Filipino Scholars in Taiwan (AFST) Taiwan
President (2021–22), Past President (2022–23), Secretary (2023–24), Treasurer (2025–Present)

- Directed organizational strategy, governance oversight, financial management, and administrative operations across multiple tenures.

ACADEMIC SERVICE

Journal Reviewer

- Applied Soft Computing (Elsevier)

- Energy Reports (Elsevier)
- Soft Computing (Elsevier)
- Discover Applied Sciences (Springer Nature)
- Applied Energy (Elsevier)
- IEEE Sensors Journal

Conference Reviewer / Program Committee

- 2026 IEEE International Conference on Systems, Man, and Cybernetics (SMC)
- 2025 IEEE Industry Applications Society Annual Meeting (IAS)

PUBLICATIONS

P1. **C. L. P. P. Rioflorido** and Y.-Y. Hong, "Adaptive Diversity-Enhanced Phasor Particle Swarm Optimization-Differential Evolution Hybrid for Unimodal-Multimodal Problems With Engineering Applications," *Machine Learning with Applications*, vol. 25, pp. 1-25, July 2026, doi: 10.1016/j.mlwa.2026.100955.

P2. Y.-Y. Hong, **C. L. P. P. Rioflorido**, Y.-Y. Wang, T.-C. Huang, S. See, H.-P. Van, M.-T. Li, "Hybrid Temporal Fusion Transformer and Quantum Long Short-Term Memory Network for Typhoon Forecasting at a Wind Power Farm," *Energy*, vol. 360, pp. 1-13, July 2026, doi: 10.1016/j.energy.2026.141900.

P3. Y.-Y. Hong, **C. L. P. P. Rioflorido**, and Y.-Y. Wang, "A Hybrid Graph-Quantum Neural Network for Solar Irradiance Forecasting Using Chaotic Swarm Optimization and Cauchy Distribution," *IEEE Transactions on Power Systems*, pp. 1619–1631, May 2026, doi: 10.1109/tpwrs.2025.3626336.

P4. Y.-Y. Hong, **C. L. P. P. Rioflorido**, and W. Zhang, "Hybrid Deep Learning and Quantum-Inspired Neural Network for Day-Ahead Spatiotemporal Wind Speed Forecasting," *Expert Systems with Applications*, vol. 241, pp. 1-11, May 2024, doi: 10.1016/j.eswa.2023.122645.

P5. Y.-Y. Hong and **C. L. P. P. Rioflorido**, "A Hybrid Deep Learning-Based Neural Network for 24-H Ahead Wind Power Forecasting," *Applied Energy*, vol. 250, pp. 530–539, Sept. 2019, doi: 10.1016/j.apenergy.2019.05.044.

SELECT CONFERENCE PROCEEDINGS

C1. Y.-Y. Hong, **C. L. P. P. Rioflorido**, C. M. Chen, C. J. Yang, J. M. F. Centeno, H. R. Limjoco, L. V. Gulmatico, M. J. N. L. Manalo, and J. L. C. Onia, "Temporal Fusion Transformer Integrated with Grey Wolf Optimizer for 24-hour Ahead Load Forecasting," in Preprints of the *23rd IFAC World Congress*, Busan, Republic of Korea, Aug. 23–28, 2026.

C2. Y.-Y. Hong, **C. L. P. P. Rioflorido**, and H.-P. Van, "Classification of Photovoltaic Faults using CatBoost Classifier Optimized by QPSO," in *The 2nd International Conference on "Green Solutions and Emerging Technologies for Sustainability" (GSETS 2025)*, Ho Chi Minh City, Vietnam, 2025, pp. 297–308.

C3. Y.-Y. Hong and C. L. P. P. Rioflorido, "A Deep CNN-LSTM Model Tuned by PPSO for Spatiotemporal Solar GHI Forecasting," in *2023 IEEE PES Innovative Smart Grid Technologies Europe (ISGT EUROPE)*, Grenoble, France, Oct. 2023, pp. 1–5, doi: 10.1109/isgteurope56780.2023.10407345.

C4. S. U. Arenas, P. J. R. Billones, J. C. V. Lao, J. A. C. Mercado, I. D. G. Sy, and **C. L. P. P. Rioflorido**, "Design of a Non-Invasive Egg Sexing Device Utilizing Artificial Intelligence for Duck Species," in *TENCON 2023 - 2023 IEEE Region 10 Conference (TENCON)*, Chiang Mai, Thailand, Oct. 31 – Nov. 3, 2023, pp. 1169–1174, doi: 10.1109/TENCON58879.2023.10322315.

C5. Y.-Y. Hong, J. B. D. Santos, S.-S. Lin, Y.-D. Lee, J.-H. Wang, J.-L. Jiang, **C. L. P. P. Rioflorido**, and J.-N. Yeh, "A Short-Term Spatiotemporal Wind Speed Forecasting Using CNN Optimized by PSO," in *2022 IET International Conference on Engineering Technologies and Applications (IET-ICETA)*, Changhua, Taiwan, Oct. 14–16, 2022, pp. 1–2, doi: 10.1109/IET-ICETA56553.2022.9971559.

C6. Y.-Y. Hong, Y.-T. Pan, C.-C. Hsu, J. B. D. Santos, and **C. L. P. P. Rioflorido**, "One-hour Ahead Spatio-Temporal Solar GHI Forecasting Using Long Short-term Memory," in *2022 IET International Conference on Engineering Technologies and Applications (IET-ICETA)*, Changhua, Taiwan, Oct. 14–16, 2022, doi: 10.1109/IET-ICETA56553.2022.9971530.

C7. D. J. A. Amora, J. V. Apan, D. N. C. Layosa, R. C. Sarmiento, J. J. Tugado, and **C. L. P. P. Rioflorido**, "Soft Sensing Measurement of Dissolved Ammonia Nitrogen in Tank-Based Eel Aquaculture Systems Utilizing Deep Learning," in *2022 IET International Conference on Engineering Technologies and Applications (IET-ICETA)*, Changhua, Taiwan, Oct. 14–16, 2022, doi: 10.1109/IET-ICETA56553.2022.9971667.

C8. D. J. A. Amora, M. A. V. Janapin, B. A. M. Calayag, **C. L. P. P. Rioflorido**, N. M. Estur, and P. A. L. Villegas, "Design of a Household Consumption Based Water Leak Detection System Utilizing Machine Learning Algorithm," in *2022 IET*

International Conference on Engineering Technologies and Applications (IET-ICETA), Changhua, Taiwan, Oct. 14–16, 2022, doi: 10.1109/IET-ICETA56553.2022.9971564.

HONORS, AWARDS, & GRANTS

- First Prize, Best Poster Competition, UST-CYCU Joint Engineering And Technology Research Symposium 2026 (September 2026)
- Grant Recipient, NVIDIA Academic Grant Program — Awarded for "Hybrid Deep Learning and Quantum Network for Typhoon Forecasting in a Wind Power Farm".
- NCHC-NVIDIA Open Hackathon Achievement (Dec 2025): optimized and achieved a 109× GPU speedup on typhoon forecasting workloads.
- Honorable Mention, CYCU × Intel × ASUS AI PC OpenVINO Application Competition (Aug 2025).
- Scholarship Recipient, CYCU Outstanding Foreign PhD Scholarship (May 2022), awarded for academic research upgrades.
- International Student Scholar, CYCU (2016, 2021).
- Honorable Mention, International Innovation Competition on Applications of Sensors & Micro-Computers (Aug 2017).